COLLEGE PROJECT · WEB APPLICATION

Web Log Analyzer

A simple browser-based tool for analyzing log files

Akshay S R

D Nithin

D Kiran

Rahul

Abhilash V

G Vinay



WHY LOGS MATTER

Logs show what happens inside an application



Events

A log file stores events created by software.



Problems

Logs help us understand normal activity and problems.



Explore

Web Log Analyzer makes these records easier to explore in a browser.

THE REVIEW PROBLEM

Large log files are hard to read manually

Slow line by line

Many records can hide important details.

Errors and repeated events can be hard to find. They are mixed in with normal records, so a reader may miss them.

Searching line by line is slow, because each record must be checked one at a time.

A simple search and filter tool makes review easier.

PROJECT GOALS

The project focuses on five simple goals

01

Make log analysis easier by running it in the browser.

02

Load JSON and JSONL log files from your computer.

03

Search and filter log records.

04

Inspect one record in detail by expanding its row.

05

Download the current analysis as a JSON report.

PROJECT TOOLKIT

Tools and file formats used in the project



HTML

HTML gives the web app its page structure.



CSS

CSS controls the page design and layout.



JavaScript

JavaScript handles user actions and runs the log analysis in the browser.



JSON / JSONL

JSON and JSONL are the two log file formats the app can load.



Python HTTP server

Runs the project on your own computer.

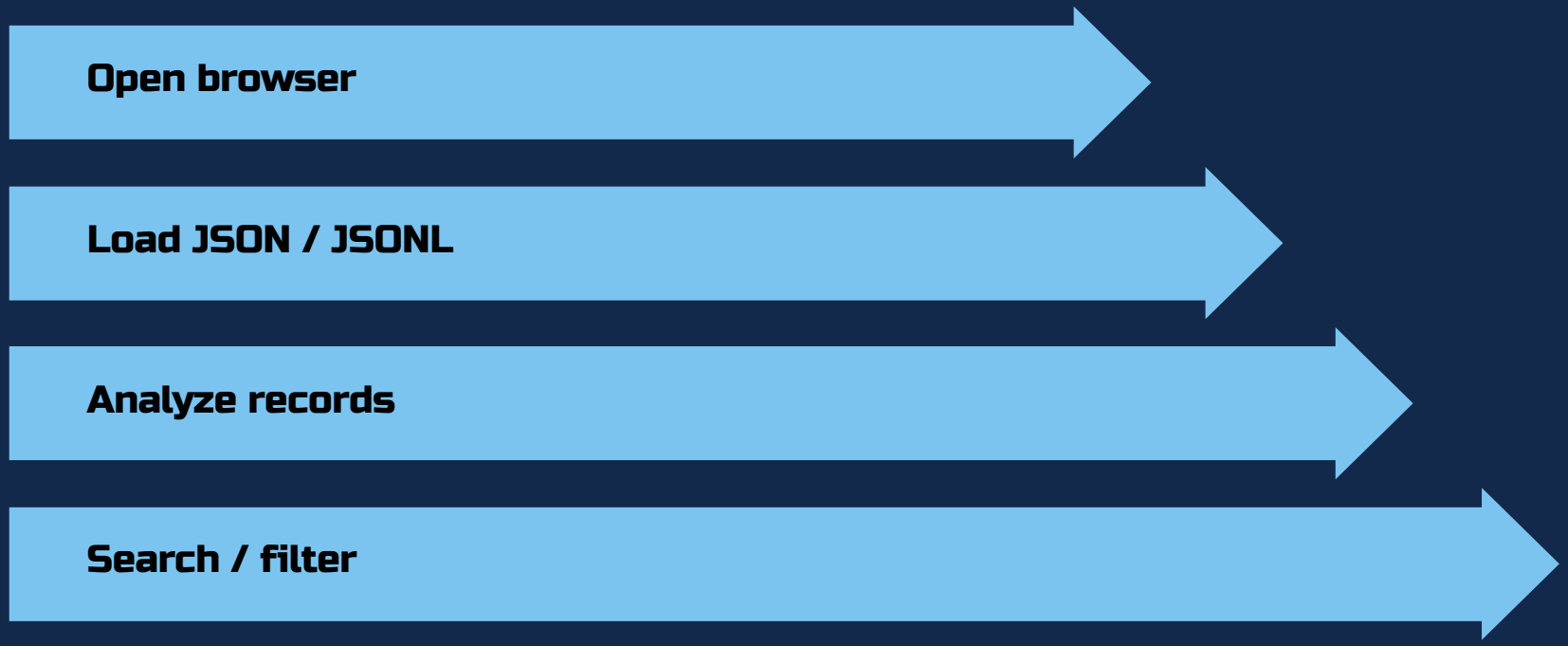


Visual Studio Code

Used to open and edit the project files.

WORKFLOW

From a local file to a useful report

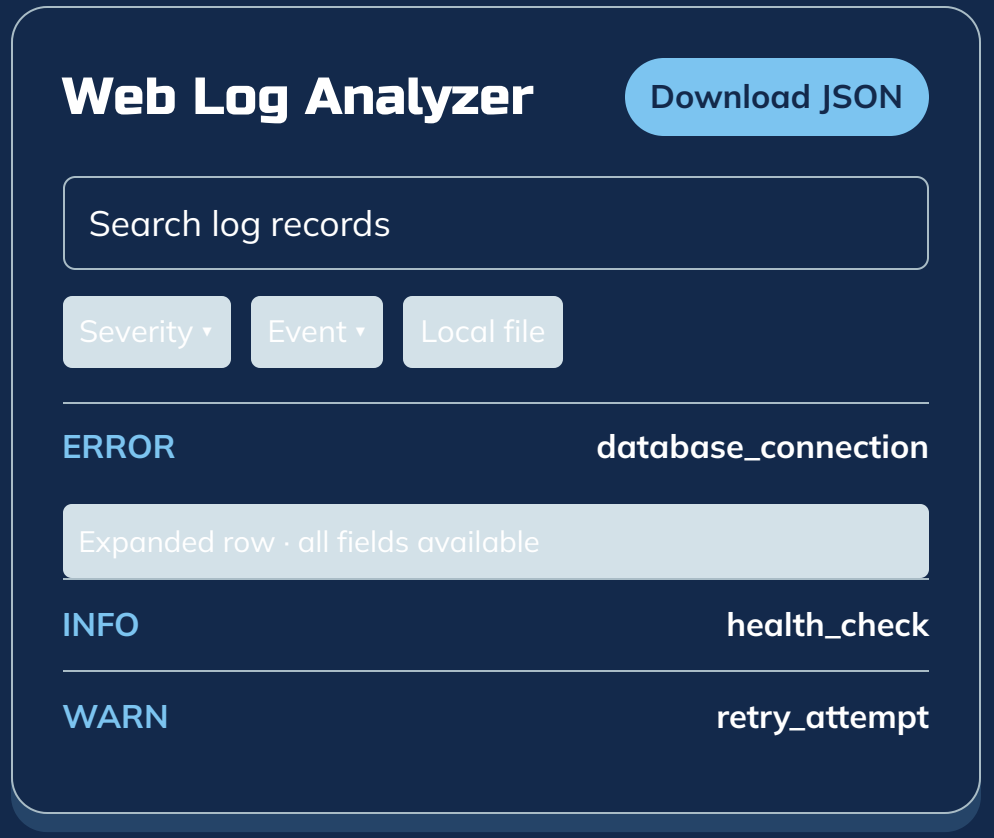


KEY FEATURES

Key features keep log review in one place

You can load a built-in sample log file, or select or drop your own JSON or JSONL file into the browser. The search box finds log records, and the severity and event filters narrow the list to the records you need.

Expand any row to see all of its fields. When your review is done, download the results as a JSON report.



WHY THIS APPROACH

Our advantage is the combination of simple choices

Each choice keeps log review direct and practical. The app runs in a web browser, analyzes files on the user's own device, and supports both JSON and JSONL.



PRACTICAL TESTING

Sample files support practical testing

The project includes four sample files. Each one tests a different part of the app, from loading a simple JSON object to grouping repeated database errors.

Sample file	What it contains	Useful for
health_check.json	One JSON object.	Basic loading
batch_jobs.json	An array of records.	Record review
api_gateway.jsonl	A malformed line and repeated errors.	Error investigation
error_burst.jsonl	Repeated database errors for testing grouping.	Grouping review

Search and filters
Help locate relevant records.

Possible uses
Debugging, log review, and error invesi

TAKEAWAY

Web Log Analyzer makes logs easier to explore.

Search, filters, record inspection, and report download support everyday review.

Local processing helps keep uploaded logs on the user’s device.

Future ideas

The project can grow next. Possible improvements include error-count charts, stronger grouping, more file formats, CSV export, and dashboard summaries.

Thank You · Any Questions?